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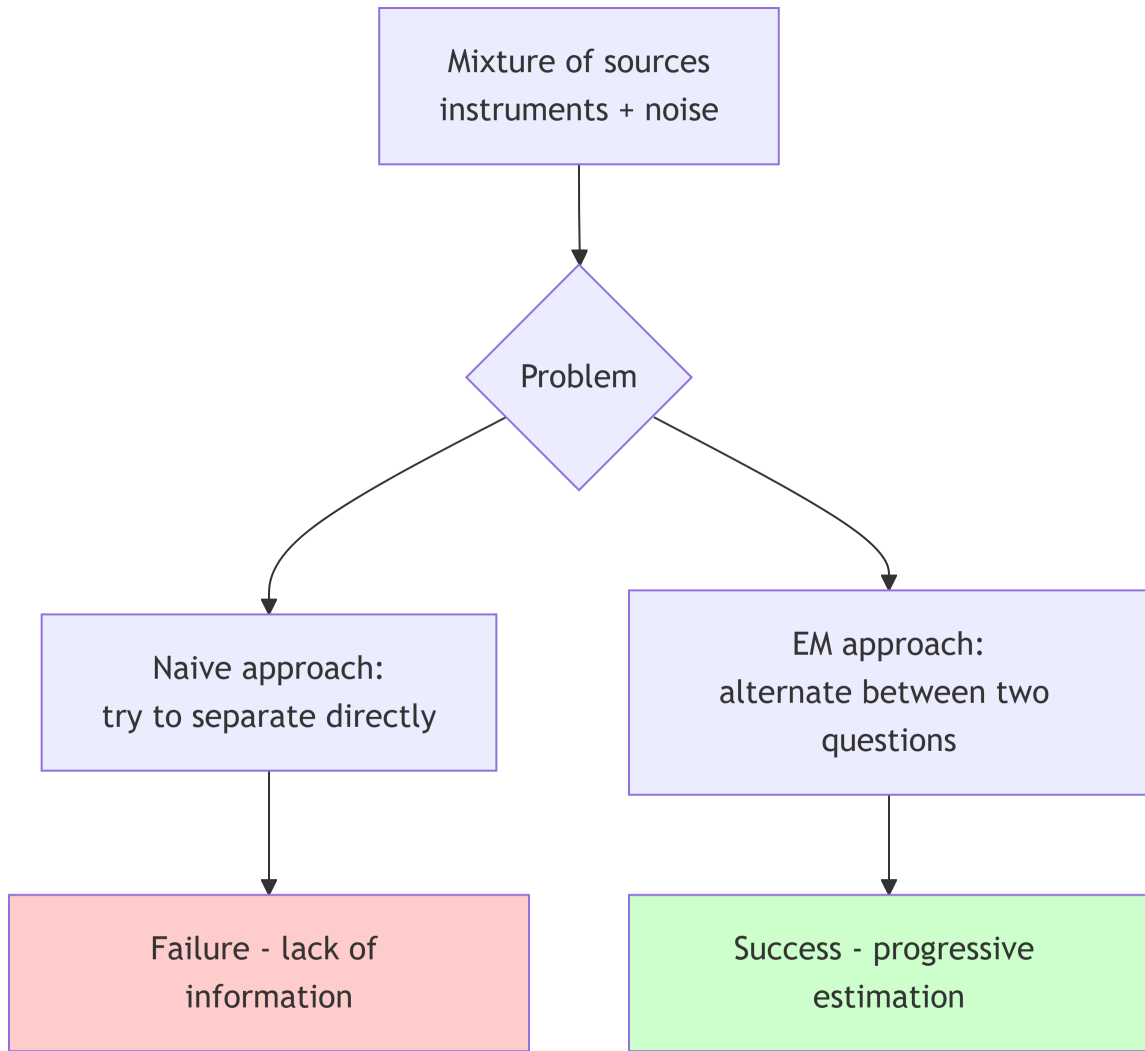
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Intuition

1. The Fundamental Problem: Separating Mixed Sources

Imagine this situation: You’re recording the sound of a room where several instruments are playing together. You have the final mix, but you want to isolate each instrument. How can you do this without knowing which note belongs to which instrument?

This is **exactly** the problem that the EM algorithm solves for Gaussian Mixture Models.



! Important

The core of the problem: With mixture models, we have **two types of unknowns** that depend on each other: 1. **The parameters** of each component (mean, variance) 2. **The origin** of each point (which component generated it)
This is a **vicious circle**: to know one, you need to know the other.

2. Gaussian Mixture Models: The Mathematics of “Landscape”

The Geometric Idea

Imagine your data forms a **mountain landscape**. Each mountain represents a natural group in your data. A Gaussian Mixture Model (GMM) represents this landscape as a **superposition of Gaussian hills**.

GMM Formula:

$$f(x) = \pi_1 \cdot N(x|\mu_1, \Sigma_1) + \pi_2 \cdot N(x|\mu_2, \Sigma_2) + \dots + \pi_K \cdot N(x|\mu_K, \Sigma_K)$$

Where: - π_j = “relative height” of hill j (mixture weight) - $N(x|\mu, \Sigma)$ = shape of the hill (Gaussian centered at μ , shape Σ)

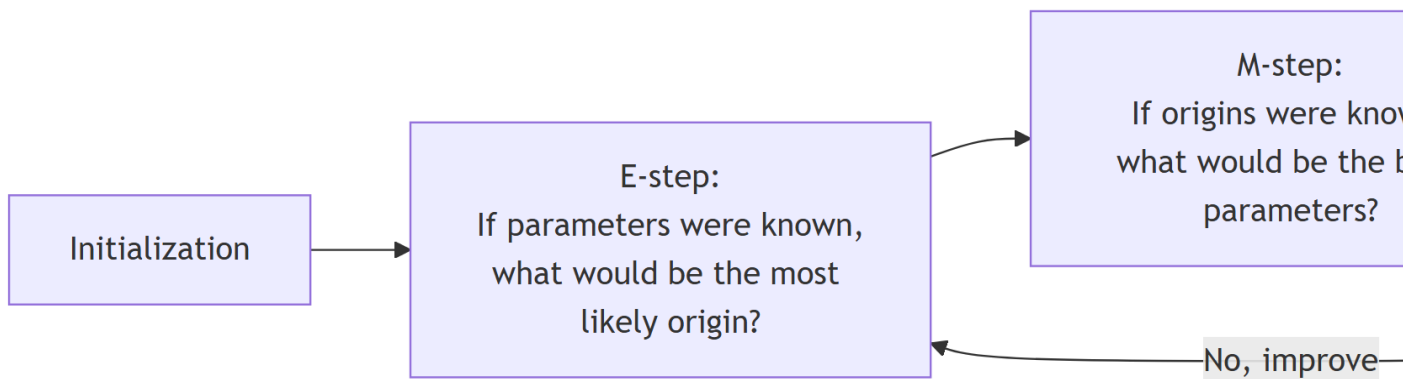
💡 Tip

Why Gaussians? 1. **Central Limit Theorem:** many natural phenomena are Gaussian 2. **Universality:** a mixture of Gaussians can approximate any density 3. **Computability:** the formulas are mathematically tractable

3. The EM Algorithm: The Two-Step Dance

The General Principle

Rather than seeking a perfect solution all at once (impossible), EM proceeds by **successive approximations** by alternating two questions:



E-step (Expectation): “Assigning Points”

Question: Given the current parameters, what is the probability that each point comes from each component?

Responsibility Calculation:

$$= P(\text{point } i \text{ comes from component } j \mid \text{current parameters}) \\ = \frac{\pi_j \cdot N(x_i \mid \mu_j, \Sigma_j)}{\sum_k \pi_k \cdot N(x_i \mid \mu_k, \Sigma_k)}$$

Interpretation: This is a **soft assignment**. Unlike K-means which says “this point is 100% in this cluster,” EM says “this point is 60% in cluster A, 30% in B, 10% in C.”

M-step (Maximization): “Updating Parameters”

Question: Given these membership probabilities, what are the best parameter estimates?

Update Formulas:

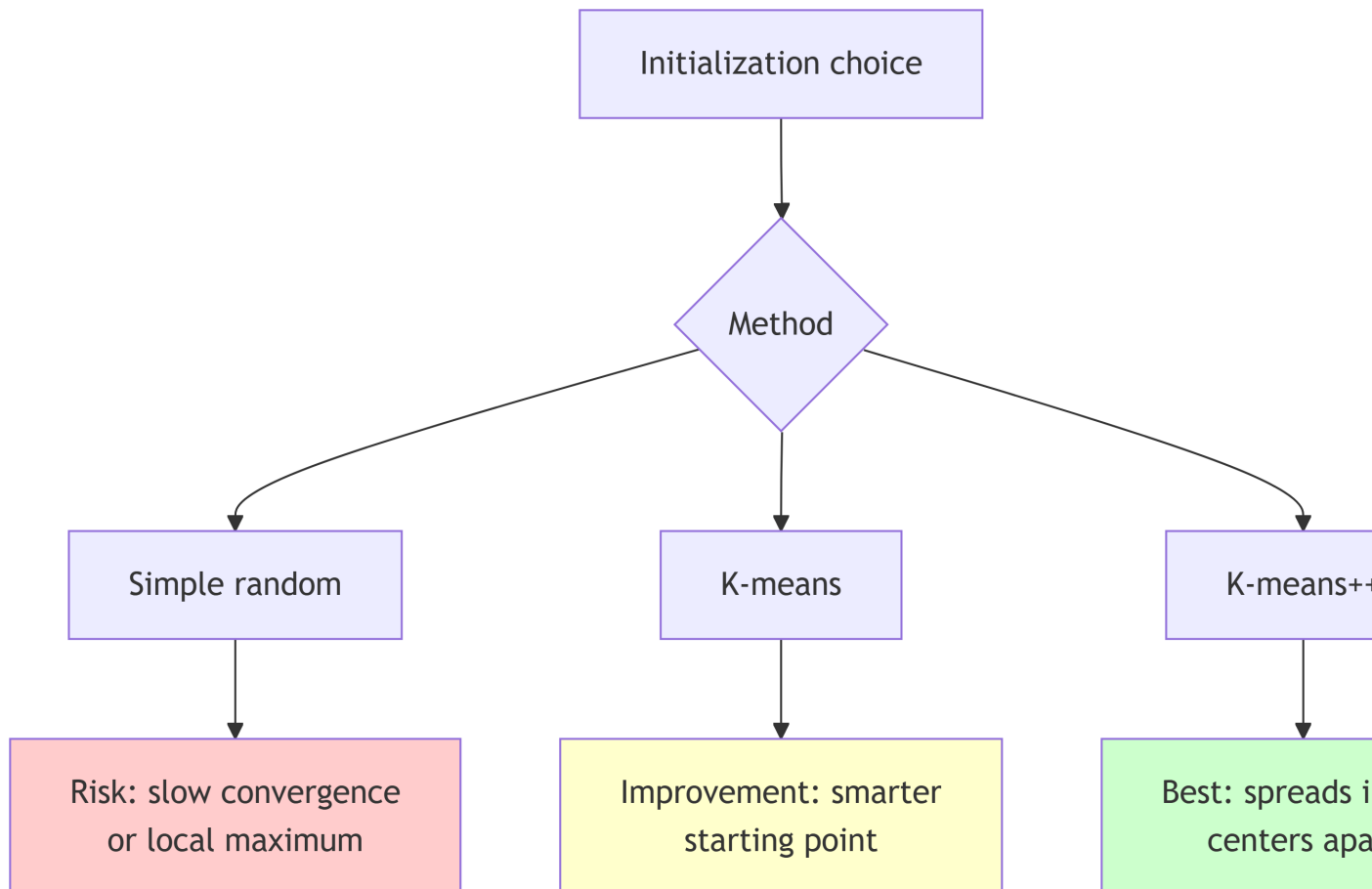
$$\begin{aligned} \mu_j &= \frac{\sum_i \pi_{ji} x_i}{\sum_i \pi_{ji}} && \text{(weighted mean)} \\ \Sigma_j &= \frac{\sum_i \pi_{ji} (x_i - \mu_j)(x_i - \mu_j)^T}{\sum_i \pi_{ji}} && \text{(weighted covariance)} \\ \pi_j &= \frac{N_j}{N} && \text{(component proportion)} \end{aligned}$$

Note

Why does it work? Each step improves the **likelihood** of the data (probability of observing this data with the current model). It’s a “**hill-climbing**” algorithm: at each iteration, we climb the likelihood hill.

4. Crucial Practical Details

Initialization: The Starting Point is Crucial



Why is this important? EM can converge to **local maxima** (small peaks) instead of the global maximum (highest peak).

Choosing K: How Many Components?

The classic dilemma: - **K too small** = underfitting = model too simple - **K too large** = overfitting = model too complex

Solution: The BIC (Bayesian Information Criterion)

$$\text{BIC} = -2 \cdot \log(\text{likelihood}) + n_{\text{parameters}} \cdot \log(n_{\text{points}})$$

Choose the **K** that minimizes **BIC** (balance between fit quality and complexity).

Covariance Matrix Form

Three main options:

Type	Form	Number of parameters	Usage
Spherical	$\Sigma = \sigma^2 \mathbf{I}$	1 per component	Isotropic data, spherical clusters
Diagonal	$\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_D^2)$	D per component	Axes aligned with principal axes
Full	Full Σ	$D(D+1)/2$ per component	Free ellipsoidal clusters

Warning

Beware of over-parameterization! With $D=256$ dimensions and $K=10$ components, a full covariance requires 330,000 parameters! Often, dimension reduction with PCA is performed first.

5. EM in Action: Complete Example

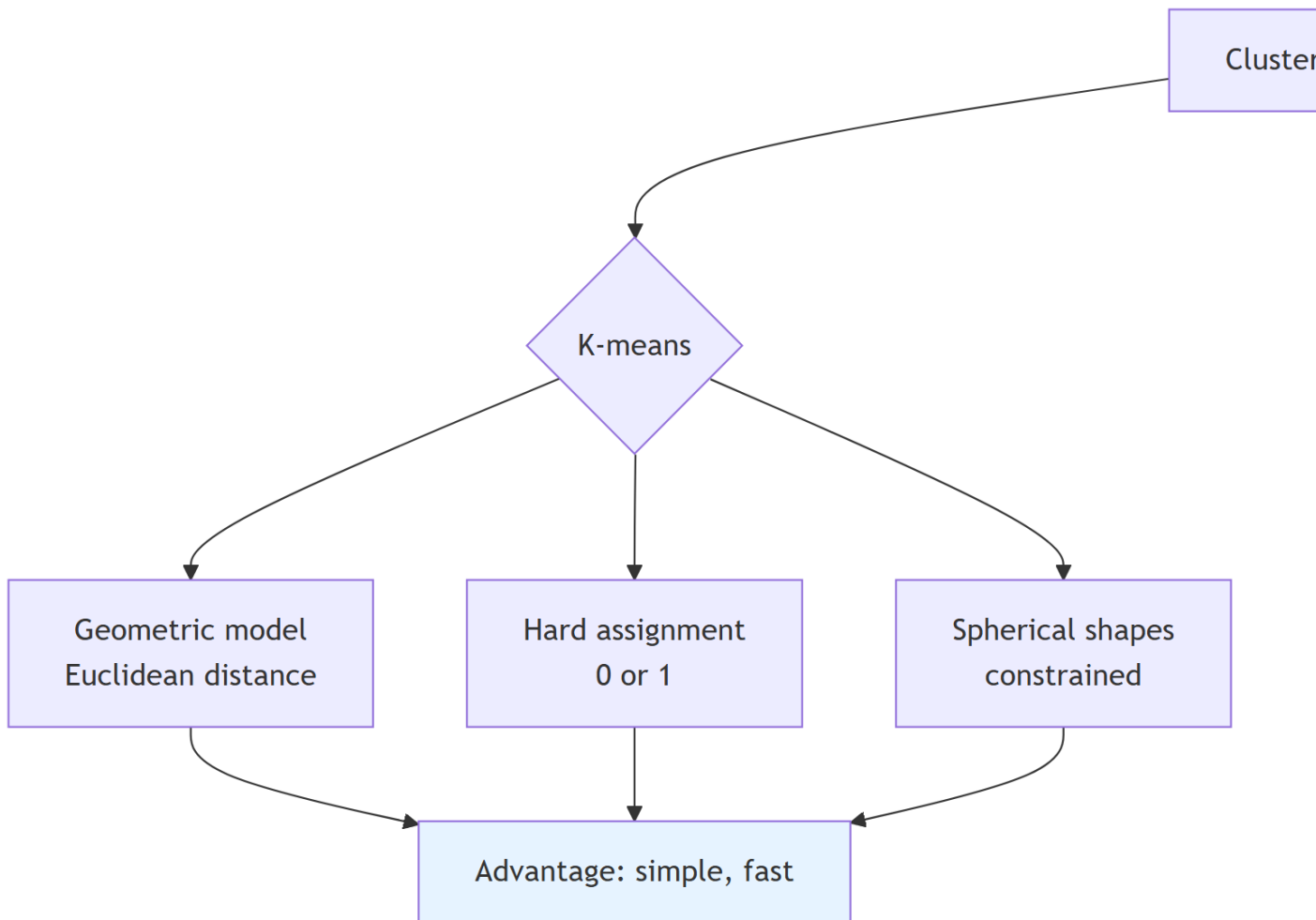
Scenario: Image Segmentation

1. **Data:** Image pixels (3 dimensions: R, G, B)
2. **Objective:** Separate the image into homogeneous regions (sky, ground, vegetation...)
3. **EM Procedure:**
 - Initialization: $K=3$ (hypothesis of 3 main regions)
 - E-step: For each pixel, calculate its probability of belonging to each region
 - M-step: Recalculate the average colors and variances of each region
 - Repeat until convergence (10-20 iterations)
4. **Result:** Probabilistic segmentation map

Convergence Visualization

Iteration 0: likelihood = -15230.4
Iteration 1: likelihood = -12456.2 (+ improvement)
Iteration 2: likelihood = -11234.5 (+ improvement)
...
Iteration 15: likelihood = -9823.1 (convergence)
Iteration 16: likelihood = -9823.0 (very small improvement → stop)

6. EM vs K-means: The Deep Difference



Key point: K-means is a **special case** of EM! When covariances are spherical and probabilities are forced to 0 or 1, we get K-means.

7. Pitfalls to Avoid and Best Practices

Pitfall 1: “Ghost Components”

Problem: EM can create components with very few points (< 10) that capture noise. **Solution:** Check the weights after convergence, merge or eliminate very small components.

Pitfall 2: Premature Convergence

Problem: The algorithm seems to converge but remains in a poor local maximum. **Solution:** Multiple trials with different initializations, monitor likelihood.

Pitfall 3: Abusive Interpretation

Problem: Thinking that the found components necessarily correspond to “real classes.” **Solution:** EM finds **statistical groups**, not necessarily semantic categories. Validate with domain knowledge.

8. Synthetic Cheat Sheet

! Important

Must remember for the oral exam:

1. **EM** = algorithm for models with latent variables
2. **Two alternating steps:** E (responsibilities) $\rightarrow$ M (parameters)
3. **GMM** = mixture of Gaussians, models data density
4. **Crucial initialization** $\rightarrow$ use K-means or K-means++
5. **Choose K** with BIC or cross-validation
6. **Covariance form:** start simple (spherical), complexify if needed
7. **Main advantage:** probabilistic model with soft assignments
8. **Relationship:** generalizes K-means (special case)
9. **Applications:** segmentation, source separation, advanced clustering
10. **Verification:** always visualize and validate results

9. Answering the Examiner's Questions

Q: “Why does EM always converge?” A: “EM guarantees that the likelihood never decreases (monotonicity). It converges to a local maximum, but not necessarily the global one.”

Q: “How to choose between EM and K-means?” A: “K-means if you want simple and fast clustering with spherical clusters. EM if you want a probabilistic model, flexible shapes, or to handle uncertainty.”

Q: “What to do if EM gives different results each time?” A: “This is normal! EM depends on initialization. You need to run it multiple times and keep the solution with the highest likelihood, or use smart initialization (K-means++).”

10. In Summary: The Philosophy of EM

The EM algorithm teaches us a profound lesson: **faced with incompleteness, iteration and alternation are powerful**. Instead of seeking the perfect solution all at once (impossible mission), we alternate between estimating causes and estimating effects, each step building on the previous one.

To conclude: EM doesn't claim to find the Truth, but the **best probabilistic explanation** given the observations and the chosen model. It's a tool of mathematical humility that recognizes and manages uncertainty.